

Robust q-negative Multifractal Detrended Cross-Correlation Coefficient

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The multifractal detrended cross-correlation coefficient $\rho_q(n)$ is widely used to investigate scale-dependent interactions, but its application to negative fluctuation orders is affected by numerical instabilities, unbounded values, and interpretational difficulties. We propose a Signed Multifractal Detrended Cross-Correlation Coefficient, $\rho_{\text{SMFDCCA}}(n, q)$, an amplitude-conditioned correlation observable for multifractal detrended analysis, based on locally normalized detrended correlations and regularized fluctuation amplitudes. The proposed coefficient preserves the sign of local interactions, remains strictly bounded within $[-1, 1]$ for both positive and negative values of q , and eliminates the corrective procedures required by previous approaches. Validation using independent fractional Gaussian noise confirms the absence of spurious cross-correlations and the numerical stability of the method. Applications demonstrate that the proposed observable resolves how cross-correlations evolve jointly with temporal scale and fluctuation amplitude, revealing scale- and amplitude-dependent correlation structures, including stronger synchronization during large fluctuations in stock-market indices and heterogeneous coupling patterns in temperature records.

INTRODUCTION

Numerous systems, composed of a large number of interacting elements organized into structures that may exist and coexist across multiple scales, exhibit emergent properties that cannot be observed in simpler systems. These elements often interact through nonlinear mechanisms, giving rise to collective behaviors and patterns that cannot be inferred from the properties of individual components [1].

A deeper understanding of the dynamics of complex time series requires the use of analytical tools that go beyond traditional approaches based on stationarity and linearity assumptions. To address these challenges, the Detrended Fluctuation Analysis (DFA) method was developed [2]. DFA has become a widely used technique for the determination of fractal scaling properties and the detection of long-range autocorrelations in noisy, nonstationary time series.

While DFA focuses on the autocorrelation of a single

signal, there are many situations where different variables are recorded simultaneously and exhibit interdependencies. To quantify long-range cross-correlations in the presence of nonstationarity, the Detrended Cross-Correlation Analysis (DCCA) was proposed [3]. DCCA is a generalization of DFA based on detrended covariance, designed to investigate power-law cross-correlations between two different time series of equal length.

However, many records require more than a single scaling exponent to be fully described, due to different scaling behaviors of small and large fluctuations. To account for this, the Multifractal Detrended Fluctuation Analysis (MFDFA) was introduced as a generalization of the DFA method. MFDFA enables the reliable multifractal characterization of nonstationary time series without requiring computationally heavy modulus maxima procedures [4].

Later, the DCCA method was also generalized to unveil the multifractal features of two cross-correlated signals. This method, known as Multifractal Detrended Cross-Correlation Analysis (MFDCCA or MF-DXA), was proposed to investigate multifractal behaviors in the power-law cross-correlations between two nonstationary time series or higher-dimensional quantities [5].

Despite the utility of DCCA and MFDCCA in identifying power-law dependencies, these methods alone did

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not directly quantify the level or strength of the cross-correlations. To solve this, was proposed the DCCA cross-correlation coefficient (ρ_{DCCA}) [6]. The ρ_{DCCA} coefficient is defined as the ratio between the detrended covariance function and the detrended variance functions. It provides a dimensionless measure ranging from -1 to 1, quantifying the level of cross-correlation in nonstationary time series, such as financial data [7, 8], mobility [9] and epidemics [10, 11].

Building upon ρ_{DCCA} method, it was proposed the multifractal cross-correlation coefficient ($\rho_q(n)$) [12]. This coefficient generalizes ρ_{DCCA} into a multifractal framework to quantify the degree of cross-correlation between two nonstationary time series across different scales as a function of the fluctuation order q . In the $\rho_q(n)$ method, the parameter q acts as a filter: for $q > 2$, it gives more weight to large fluctuations, whereas $q < 2$ emphasizes small fluctuations, allowing researchers to map whether cross-correlations are homogeneous across all amplitudes or dominated by specific fluctuation magnitudes.

Recently, was proposed the Multifractal detrended multiple cross-correlation coefficient ($qDMC_x^2(n)$) [13]. This coefficient constructs a methodology by integrating the multifractal cross-correlation coefficient $\rho_q(n)$ with the standard multiple correlation coefficient $DMC_x^2(n)$.

But a critical issue associated with multifractal detrended cross-correlation methods emerges for negative values of the fluctuation order parameter $q < 0$. In multifractal analysis, negative q -orders amplify the contribution of segments characterized by very small fluctuation amplitudes, while positive q -orders emphasize large fluctuations. Although this property is theoretically useful for investigating weak fluctuations, it introduces important mathematical and numerical instabilities.

According to the discussion presented by Kwapien et al. [12], the main difficulty associated with negative values of the fluctuation order parameter ($q < 0$) arises from the excessive emphasis placed on segments containing very small fluctuations. While positive values of q highlight large-amplitude fluctuations, negative values strongly amplify low-amplitude regions, making the coefficient extremely sensitive to weak signals, noise, and numerical inaccuracies. As a consequence, the multifractal cross-correlation coefficient may lose its conventional interpretation as a bounded correlation measure, since values larger than unity in magnitude can emerge for $q < 0$. Kwapien et al. explicitly describe the interpretation of negative- q results as a delicate issue, emphasizing that the coefficient may exhibit unstable and erratic behavior in this regime. To reduce this problem, they introduced an additional inversion procedure whenever the coefficient exceeds the expected bounds. Therefore, the original qDCCA formulation does not intrinsically resolve the instability associated with negative fluctuation orders, but rather compensates for it through a corrective post-processing step.

The same limitation propagates directly into the recently proposed multifractal detrended multiple cross-correlation coefficient $qDMC_x^2(n)$ developed by Silva Filho et al. [13], since the method is constructed explicitly from the original $\rho_q(n)$ coefficient. Consequently, the numerical instability and boundedness problems associated with $q < 0$ remain embedded in the multivariate framework. It is important to note that Sierra-Porta [14], in the analysis of cosmic ray intensity and sunspot number using the multifractal detrended cross-correlation coefficient $\rho_q(n)$, restricted the investigation to positive fluctuation orders ($q \geq 0$). By excluding negative q -values, Sierra-Porta avoided these instability issues and restricted the analysis to fluctuation regimes where the coefficient remains more numerically robust and physically interpretable.

Despite the advances provided by the multifractal detrended cross-correlation coefficient $\rho_q(n)$ and its multivariate extension $qDMC_x^2(n)$, some mathematical and numerical limitations remain unresolved, particularly for negative fluctuation orders ($q < 0$). Such difficulties become relevant when studying systems characterized by weak fluctuations, intermittency, or near-constant local segments. Motivated by these limitations, the present work introduces a amplitude-conditioned correlation observable for multifractal detrended analysis, namely the Signed Multifractal Detrended Cross-Correlation Coefficient, based on locally normalized signed correlations and regularized fluctuation amplitudes. In the present context, the term multifractal refers to the q -dependent weighting of fluctuation amplitudes, which enables the characterization of cross-correlations associated with different fluctuation magnitudes. The proposed observable provides a bounded measure of amplitude-conditioned cross-correlations across temporal scales for both positive and negative values of q .

SIGNED MULTIFRACTAL DETRENDED CROSS-CORRELATION COEFFICIENT

Let x_i and y_i , with $i = 1, \dots, N$, be two time series of equal length. The integrated profiles are first constructed as

$$X_k = \sum_{i=1}^k (x_i - \bar{x}), \quad Y_k = \sum_{i=1}^k (y_i - \bar{y}), \quad k = 1, \dots, N. \quad (1)$$

The profiles are divided into N_n segments of equal length n . In each segment v , polynomial trends of order m are fitted to the profiles:

$$\tilde{X}_{v,m}(k), \quad \tilde{Y}_{v,m}(k), \quad k = 1, \dots, n. \quad (2)$$

The detrended residuals are then obtained as

$$\varepsilon_{x,v}(k) = X_v(k) - \tilde{X}_{v,m}(k), \quad (3)$$

$$\varepsilon_{y,v}(k) = Y_v(k) - \tilde{Y}_{v,m}(k). \quad (4)$$

The local detrended variances are defined by

$$F_x^2(n, v) = \frac{1}{n} \sum_{k=1}^n \varepsilon_{x,v}^2(k), \quad (5)$$

$$F_y^2(n, v) = \frac{1}{n} \sum_{k=1}^n \varepsilon_{y,v}^2(k). \quad (6)$$

The local detrended covariance is

$$C_{xy}(n, v) = \frac{1}{n} \sum_{k=1}^n \varepsilon_{x,v}(k) \varepsilon_{y,v}(k). \quad (7)$$

Unlike the local variances, $C_{xy}(n, v)$ may assume positive or negative values depending on the direction of the local cross-correlation.

To avoid numerical instabilities and division by zero, particularly for negative values of q , a small positive regularization constant $\epsilon > 0$ is introduced. This parameter acts as a numerical floor that prevents singularities and undefined operations when the detrended variances become extremely small or vanish within a local segment. The local fluctuation amplitude is then defined as

$$A_v(n) = \sqrt{\max(F_x^2(n, v), \epsilon) \max(F_y^2(n, v), \epsilon)}. \quad (8)$$

The regularization is also essential for the evaluation of logarithmic and power-law weighting terms, especially for negative multifractal orders $q < 0$, where small fluctuation amplitudes receive disproportionately large weights. Without regularization, segments with vanishing amplitudes would lead to weights and numerical instabilities. The parameter ϵ is therefore chosen as a very small constant without loss of generality ($\epsilon = 10^{-12}$), so that it remains negligible compared to the characteristic fluctuation amplitudes of the data while still guaranteeing mathematically well-defined and computationally stable calculations. Although ϵ slightly regularizes the behavior of extremely low-amplitude segments, its influence on the overall multifractal structure is minimal for practical ranges of q .

Using this quantity, the local detrended correlation coefficient becomes

$$r_v(n) = \frac{C_{xy}(n, v)}{A_v(n)}. \quad (9)$$

By the Cauchy-Schwarz inequality,

$$-1 \leq r_v(n) \leq 1. \quad (10)$$

For a multifractal order $q \in \mathbb{R}$, the weight assigned to each segment is

$$w_v(q, n) = A_v(n)^{q/2}, \quad q \neq 0. \quad (11)$$

For $q = 0$, the limiting case corresponds to equal weighting of all segments:

$$w_v(0, n) = 1. \quad (12)$$

The signed multifractal detrended cross-correlation coefficient is then defined as

$$\rho_{\text{SMFDCCA}}(n, q) = \frac{\sum_{v=1}^{N_n} w_v(q, n) r_v(n)}{\sum_{v=1}^{N_n} w_v(q, n)}, \quad q \in \mathbb{R}. \quad (13)$$

Equation (13) defines an amplitude-conditioned correlation observable obtained as the expectation of bounded local detrended correlation coefficients under a q -dependent weighting of fluctuation amplitudes. Consequently, each value of q probes a distinct fluctuation-amplitude regime while preserving the interpretation of the observable as a bounded signed correlation coefficient. Equivalently,

$$\rho_{\text{SMFDCCA}}(n, q) = \frac{\sum_{v=1}^{N_n} A_v(n)^{q/2} \frac{C_{xy}(n, v)}{A_v(n)}}{\sum_{v=1}^{N_n} A_v(n)^{q/2}}, \quad q \neq 0. \quad (14)$$

For $q = 0$, the coefficient reduces to the arithmetic mean of the local detrended correlations:

$$\rho_{\text{SMFDCCA}}(n, 0) = \frac{1}{N_n} \sum_{v=1}^{N_n} r_v(n). \quad (15)$$

Since the weights satisfy

$$w_v(q, n) \geq 0, \quad (16)$$

the coefficient is a weighted average of local correlations and therefore satisfies

$$-1 \leq \rho_{\text{MFDDCCA}}(n, q) \leq 1. \quad (17)$$

This formulation is valid for positive, negative, and zero values of q . Positive values of q emphasize segments with larger fluctuation amplitudes, while negative values emphasize smaller fluctuations. Consequently, $\rho_{\text{SMFDCCA}}(n, q)$ can be interpreted as the expected local detrended correlation conditioned by fluctuation amplitude, providing a physically interpretable observable across different fluctuation regimes. Because the sign

of the local covariance $C_{xy}(n, v)$ is preserved through $r_v(n)$, the coefficient simultaneously quantifies both the strength and direction of multifractal cross-correlations.

The $\rho_{\text{SMFDCCA}}(n, q)$ coefficient is rooted in the same q -dependent fluctuation formalism that underlies MFCCA and $\rho_q(n)$. As discussed by Wątorrek *et al.* [15], q -dependent correlation coefficients and multifractal cross-correlation exponents provide complementary information about the interaction structure of complex systems, with the parameter q acting as a selective filter for fluctuation amplitudes. By preserving this fluctuation-selective weighting mechanism, $\rho_{\text{SMFDCCA}}(n, q)$ retains the ability to distinguish correlation structures associated with different fluctuation magnitudes.

It is important to emphasize that the multifractal character of $\rho_{\text{SMFDCCA}}(n, q)$ arises from its explicit dependence on the fluctuation order q , which defines a family of amplitude-conditioned observables. Consequently, the coefficient provides an amplitude-stratified characterization of cross-correlations. Unlike multifractal scaling frameworks such as MFDFA or MFCCA, the proposed coefficient is not designed to estimate generalized Hurst exponents, multifractal spectra, or singularity distributions. Its purpose is to quantify how the strength and direction of cross-correlations vary across fluctuation magnitudes and temporal scales while preserving boundedness and numerical stability.

DISCUSSION

Computer Generated Random Signal

To establish a baseline for the proposed observable, we generated two independent fractional Gaussian noise time series using the exact Davies–Harte method. Each series consists of $N = 20,000$ data points characterized by a Hurst exponent $H = 0.5$, indicative of independent stochastic processes devoid of long-range auto-correlations. The logarithmic returns of both series were subsequently computed to evaluate the background noise response of the $\rho_{\text{SMFDCCA}}(n, q)$ coefficient (see Figure 1).

As established in the multiscale analysis literature, simulating independent Gaussian processes provides a null-model validation for evaluating detrended cross-correlation estimators [3, 12, 13]. Throughout the investigated range $q \in [-10, 10]$, the $\rho_{\text{SMFDCCA}}(n, q)$ coefficient fluctuates around zero across all temporal scales n , with average values on the order of 10^{-3} and maximum absolute deviations strictly bounded below 5×10^{-3} . This asymptotic decoupling is expected for independent random walks ($H = 0.5$), confirming that no spurious cross-correlations are falsely detected across any scale or fluctuation amplitude [3].

The minor residual oscillations around zero are thoroughly consistent with well-documented finite-sample effects and detrending inefficiencies at specific scales,

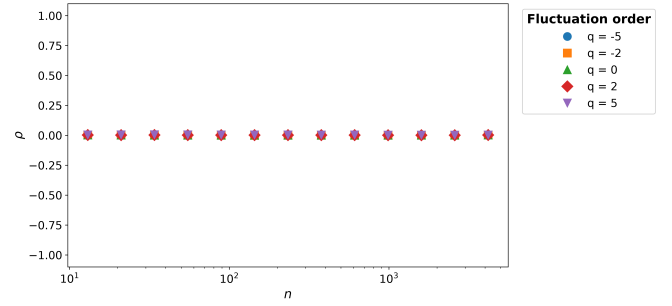


FIG. 1. Evaluation of $\rho_{\text{SMFDCCA}}(n, q)$, using two independent fractional Gaussian noise time series generated by the exact Davies–Harte method with Hurst exponent $H = 0.5$ and 20,000 samples each. For all investigated scales n and fluctuation orders $q \in [-10, 10]$, the coefficient remains close to zero, indicating the absence of cross-correlations between the signals. The residual fluctuations around zero are consistent with finite-sample effects and demonstrate the numerical stability of the proposed coefficient for both positive and negative values of q .

rather than any underlying dynamical interaction [4, 12]. The absence of divergence in the negative q regime empirically demonstrates the numerical stability of our signed formulation, overcoming the instability limitations historically associated with extracting cross-correlations from small-amplitude fluctuations.

U.S. equity markets

To empirically validate the proposed observable, we evaluate the amplitude-conditioned correlation $\rho_{\text{SMFDCCA}}(n, q)$ coefficient on the daily logarithmic returns of the Dow Jones Industrial Average (^DJI) and NASDAQ Composite (^IXIC) indices from January 2, 1992, to May 8, 2026 ($N = 8649$), collected from Yahoo Finance. The presence of power-law cross-correlations between these two benchmark U.S. markets is a well-established stylized fact in the complex systems literature [3, 6]. As illustrated in Figure 2, our analysis reveals persistent positive cross-correlations across all investigated temporal scales n and fluctuation orders $q \in [-5, 5]$, with the coefficient bounded between 0.49 and 0.93. We observe a pronounced asymmetry in the correlation structure: positive moments ($q > 0$) yield significantly stronger cross-correlations than negative moments ($q < 0$). This discrepancy explicitly confirms that large-amplitude fluctuations are characterized by a much higher degree of synchronization than small-amplitude fluctuations, a phenomenon consistent with the non-uniform coupling typically observed in empirical multifractal systems [12, 16].

A scale dependence is also observed. At short windows, the cross-correlation spectrum exhibits the largest spread, increasing from $\rho_{\text{SMFDCCA}}(13, -5) = 0.486$ to $\rho_{\text{SMFDCCA}}(13, 5) = 0.931$. It may indicate a significant

heterogeneity in the q -dependent coupling mechanism, where large return fluctuations are almost synchronized while small fluctuations remain only moderately synchronized [12, 13]. As the scale increases, the dependence on q progressively weakens. For the largest investigated window ($n = 1597$), the coefficient varies only from 0.782 to 0.854, revealing a substantial reduction of the amplitude-dependent asymmetry in the correlation structure. This convergence suggests that long-term market dynamics are dominated by common macroscopic factors that homogenize the correlation structure across fluctuation amplitudes. It is consistent with the presence of nonlinear interactions and scale-dependent coupling mechanisms, which are generally found on complex financial systems [16–18].

The persistence of strong positive ρ_q values demonstrates multiscale co-movements between the industrial and technology sectors. By deploying the amplitude-conditioned correlation observable $\rho_{\text{SMFDCCA}}(n, q)$, we can quantitatively present how this interaction strength evolves simultaneously with both temporal scale and fluctuation magnitude. These findings underscore the nonlinear, hierarchical organization of modern financial markets [16, 19], proving the efficacy of the proposed observable in characterize multiscale co-movements. $\rho_{\text{SMFDCCA}}(n, q)$ may also provide an amplitude-resolved view of the scale-dependent synchronization traditionally associated with the Epps effect [20, 21].

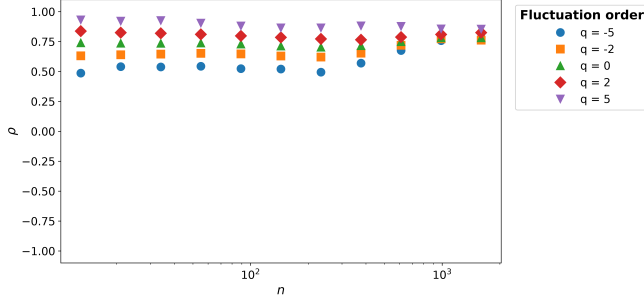


FIG. 2. $\rho_{\text{SMFDCCA}}(n, q)$ between the daily logarithmic returns of the Dow Jones Industrial Average and NASDAQ Composite Index from January 2, 1992 to May 8, 2026 ($N = 8649$). The coefficient is shown for fluctuation orders $q \in [-5, 5]$ as a function of the window size n . Positive values of $\rho_{\text{SMFDCCA}}(n, q)$ across all scales indicate persistent cross-correlations between the two U.S. equity markets. The increase of $\rho_{\text{SMFDCCA}}(n, q)$ with q reveals strong synchronized co-movements for large-amplitude fluctuations than for small fluctuations, evidencing q -dependent asymmetry in the correlation structure.

Weather data

The weather graphs displayed in Figure 3 reveal different cross-correlation structures depending on the climatic variable pair. For the mean temperature–mean humidity relationship, São Paulo (Fig. 3a) exhibit strong

negative cross-correlations at short temporal scales, indicating that local increases in temperature are generally associated with decreases in relative humidity. The coefficient evolves from strongly negative values at short scales toward positive values at intermediate scales before partially reverting at the largest windows, suggesting a complex dynamic between local atmospheric processes and seasonal forcing.

The maximum temperature–minimum temperature pairs (Fig. 3b) exhibit a different co-movement behavior. $\rho_{\text{SMFDCCA}}(n, q)$ remains positive across almost all scales and fluctuation orders, indicating persistent coupling between daily temperature measurements. A strong co-movement observed in São Paulo, where the coefficient approaches unity at intermediate and large scales, indicating a very strong coupling between maximum and minimum temperatures. Positive fluctuation orders produce larger coefficients than negative orders, indicating that large temperature fluctuations are more strongly synchronized than weak fluctuations.

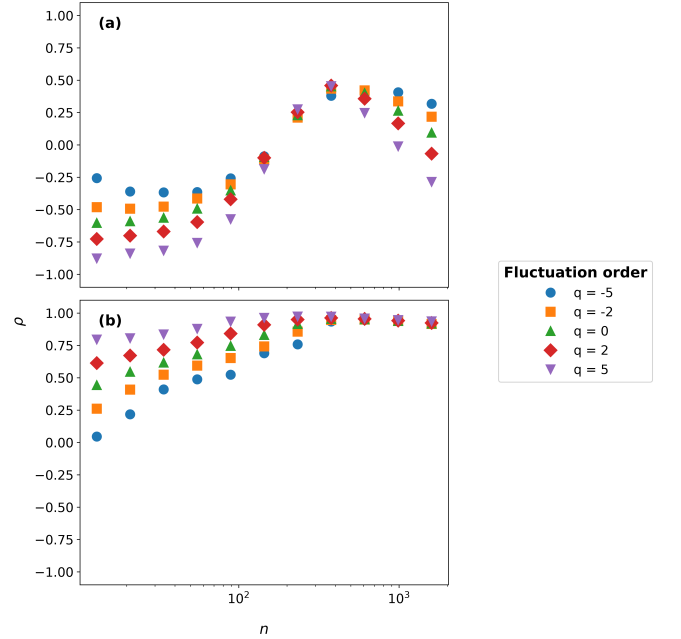


FIG. 3. $\rho_{\text{SMFDCCA}}(n, q)$, for meteorological variables in São Paulo, Brazil. (a) Mean temperature versus mean relative humidity in São Paulo. (b) Maximum versus minimum temperature in São Paulo. The dependence on the fluctuation order $q \in [-5, 5]$ reveals heterogeneous cross-correlation structures across scales.

Limitations

Although the proposed observable is mathematically bounded and preserves the sign of local cross-correlations, extreme positive and negative values of q may induce numerical and statistical instabilities. The overall cross-correlation coefficient is derived from a

weighted average of local cross-correlations r_i . The weight W_i of each local segment i is proportional to its fluctuation amplitude raised to the power of q :

$$W_i \propto (\text{Amplitude}_i)^q \quad (18)$$

This is the reason the method is sensitive to localized variance spikes when evaluated at these high moment orders, particularly at large temporal scales where the total number of segments is inherently small. Consequently, practical applications may employ moderate q -ranges.

For daily or low-frequency financial return series, empirical guidelines suggest restricting the range of q to $|q| \leq 6$, or even smaller, to preserve estimation accuracy [16]. When dealing with heavy-tailed datasets such as high-frequency equity or cryptocurrency returns (where the tail index $\gamma \approx 3$), the range of q is typically restricted to $[-4, 4]$ due to the non-existence of higher-order moments in the underlying probability density functions [15]. Furthermore, for certain polynomial or parametric fits of scaling functions, researchers often restrict the analysis to $q \geq -1$ because when the empirical tail index satisfies $\gamma > 2$, moments of returns for $q < -1$ do not formally exist [16].

SUMMARY

This work introduces the Signed Multifractal Detrended Cross-Correlation Coefficient, $\rho_{\text{SMFDCCA}}(n, q)$, the first amplitude-conditioned correlation observable within the multifractal detrended analysis framework, for quantifying multifractal cross-correlations while preserving both the strength and direction of local interactions. Unlike previous multifractal cross-correlation coefficients, the proposed formulation is based on locally normalized detrended correlations and positive-definite fluctuation weights, resulting in a bounded coefficient within the interval $[-1, 1]$ for all values of the fluctuation order q . The method overcomes a longstanding limitation of conventional q -dependent detrended cross-correlation approaches, namely the numerical instability and loss of interpretability associated with negative fluctuation orders. By introducing a regularized local fluctuation amplitude and performing the averaging directly on bounded local correlation coefficients, the proposed approach eliminates the need for corrective inversion procedures and provides a unified treatment of positive and negative moments.

The proposed observable was validated through a combination of synthetic and empirical applications. For independent fractional Gaussian noise series generated with $H = 0.5$, the coefficient remained close to zero for all scales and fluctuation orders, confirming the absence of spurious cross-correlations and demonstrating numerical robustness in both positive and negative q regimes. Application to the daily logarithmic returns of the Dow Jones Industrial Average and NASDAQ Composite indices revealed persistent positive cross-correlations

across all investigated scales, with stronger synchronization for large-amplitude fluctuations than for small fluctuations. These findings demonstrate that the proposed coefficient successfully captures scale-dependent and amplitude-dependent coupling structures while preserving a straightforward interpretation as a bounded signed correlation measure. More generally, the results indicate that the strength of cross-correlations may vary substantially across fluctuation magnitudes, highlighting the importance of amplitude-stratified analyses in complex systems.

The analysis of meteorological records further highlighted the ability of $\rho_{\text{SMFDCCA}}(n, q)$ to uncover heterogeneous coupling mechanisms that remain hidden to conventional correlation estimators. Such results demonstrate the capacity of the proposed observable to present amplitude-conditioned cross-correlation structures that remain hidden to conventional correlation estimators. More generally, the proposed observable extends the concept of detrended cross-correlation by introducing fluctuation amplitude as an explicit conditioning variable, thereby providing a complementary description of multiscale interactions that is inaccessible to conventional correlation observables.

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END MATTER

Numerical Stability of the Regularization Parameter

The computation of the Signed Multifractal Detrended Cross-Correlation Coefficient employs a small positive regularization parameter, ε , exclusively to prevent numerical singularities when evaluating negative fluctuation orders ($q < 0$). The regularization is introduced only to ensure numerical robustness when local detrended fluctuation amplitudes approach zero and does not modify the mathematical definition of the proposed coefficient.

To evaluate the numerical sensitivity of the implementation, $\rho_{\text{SMFDCCA}}(n, q)$ was recomputed for

$$\varepsilon \in \{10^{-20}, 10^{-18}, 10^{-16}, 10^{-14}, 10^{-12}, 10^{-10}, 10^{-8}, 10^{-6}\},$$

using all application datasets presented in this work, namely the Dow Jones Industrial Average versus NASDAQ Composite, mean temperature versus mean relative humidity in São Paulo, and maximum versus minimum temperature in São Paulo.

For each dataset, the complete $\rho_{\text{SMFDCCA}}(n, q)$ surface was recomputed using the representative fluctuation orders

$$q = -5, -2, 0, 2, 5,$$

and compared with the reference implementation using $\varepsilon = 10^{-12}$. The Root Mean Square Error (RMSE), maximum absolute deviation, and mean deviation were evaluated for every tested value of ε .

No measurable numerical differences were observed over the entire investigated interval ($10^{-20} \leq \varepsilon \leq 10^{-6}$). All recomputed coefficients were numerically indistinguishable from the reference implementation, with zero observed RMSE and zero maximum deviation across all datasets and fluctuation orders.

These results demonstrate that the proposed coefficient possesses a broad numerical stability plateau and is effectively insensitive to the precise choice of the regularization parameter over fourteen orders of magnitude. Consequently, $\varepsilon = 10^{-12}$ was adopted as the default value throughout this work.

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